CS 4420 Assignment 06

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1. Overview

This program implements a majority voting system with one judge and N voters, where N is chosen by the user (N ≥ 3). The judge is the parent process and each voter is a child process created with fork(). In every round each voter randomly chooses YES or NO, prints its vote, and sends it to the judge through a pipe. The judge collects the N votes, applies the majority rule, and prints the result. The same group of voters takes part in every round, and all processes terminate cleanly after the last round.

2. Process Interaction Diagram

The judge creates one pipe per voter. Voter i writes only to pipe i, and the judge reads from every pipe. Communication is one-directional (voter → judge).

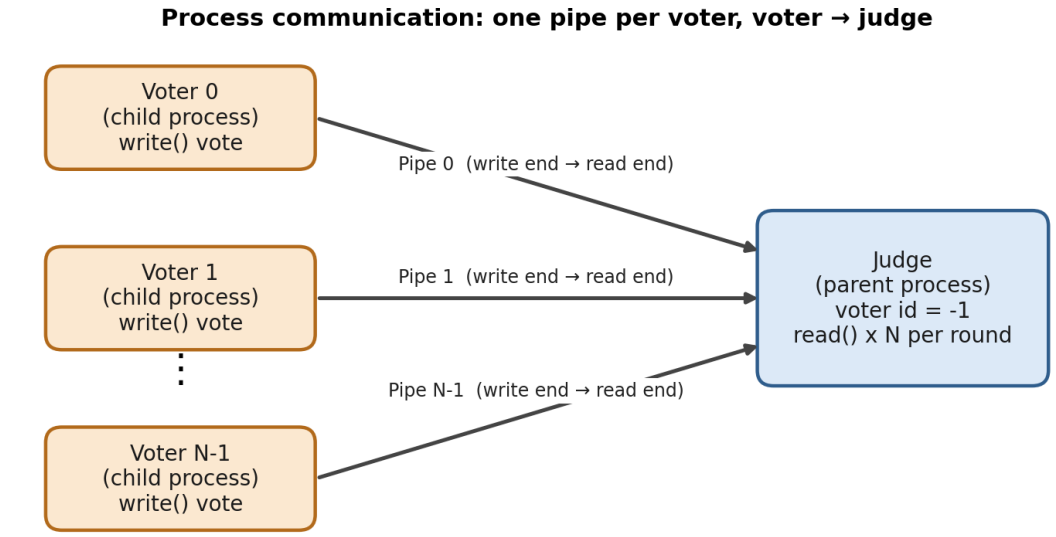


Figure 1: Judge and voter processes connected by N pipes

3. Program Design

Program steps

1. The program asks for the number of voters N and the maximum number of rounds, and rejects N < 3.
2. The judge creates N pipes with pipe() in a for loop. Pipe i connects Voter i to the judge.
3. The judge calls fork() N times in a second for loop. Each child records its loop index as its voter id and stops forking, while the judge keeps voter id = -1. Only the judge forks, and no process exits at this stage.

4. For each round, the judge reads one vote from each pipe (inner for loop) and prints the majority result. Each voter generates a random vote, prints it, and writes it to its own pipe with write(), with no loop.
5. Cleanup: Each process closes the pipe ends it does not use. Voters exit after the last round, and the judge calls wait() once per child so that no zombie processes remain.

4. How to Compile and Run

The program is written in C and was built with gcc

Compile:

```
gcc -Wall -o Assignment6 Assignment6.c
```

Run:

```
./Assignment6
```

Enter the number of voters (at least 3) and the number of rounds when prompted.

5. Sample Runs

Run 1: 3 voters, 2 rounds

```
jacobr@LAPTOP-P4Q45RUC:/mnt/c/Users/jake5/.vscode/C++ code/.vscode/operating_systems/Assignment6$ gcc -Wall -o Assignment6 Assignment6.c
jacobr@LAPTOP-P4Q45RUC:/mnt/c/Users/jake5/.vscode/C++ code/.vscode/operating_systems/Assignment6$ ./Assignment6
Enter the number of voters: 3
Enter the number of rounds: 2
Round 1: Voter 0 votes: YES
Round 2: Voter 0 votes: YES
Round 1: Voter 1 votes: NO
Round 2: Voter 1 votes: YES

------ Round 1 -------
Round 1: Voter 2 votes: NO
Round 2: Voter 2 votes: NO
--- Round 1: The voting result is a NO.

------ Round 2 -------
--- Round 2: The voting result is a YES.
```

Figure 2: Screenshot of Run 1 (3 voters, 2 rounds)

Verification of Run 1 (3 voters, 2 rounds)

The majority result of each round can be checked against the printed votes:

- **Round 1:** YES, NO, NO gives 1 YES and 2 NO, so the result is NO.
- **Round 2:** YES, YES, NO gives 2 YES and 1 NO, so the result is YES.

The voter lines are not in strict round order. Voters 0 and 1 printed both of their votes before the judge printed the Round 1 header. This happens because each voter runs independently and does not wait for the judge.

Run 2: 5 voters, 3 rounds

```
Enter the number of voters: 5
Enter the number of rounds: 3
Round 1: Voter 0 votes: YES
Round 2: Voter 0 votes: NO
Round 3: Voter 0 votes: YES
Round 1: Voter 1 votes: NO
Round 2: Voter 1 votes: YES
Round 3: Voter 1 votes: YES
Round 1: Voter 2 votes: YES
Round 2: Voter 2 votes: YES
Round 3: Voter 2 votes: NO
Round 1: Voter 3 votes: NO
Round 2: Voter 3 votes: NO
Round 3: Voter 3 votes: NO

------ Round 1 -------
Round 1: Voter 4 votes: YES
Round 2: Voter 4 votes: YES
--- Round 1: The voting result is a YES.
Round 3: Voter 4 votes: NO

------ Round 2 -------
--- Round 2: The voting result is a YES.

------ Round 3 -------
--- Round 3: The voting result is a NO.
```

Figure 3: Screenshot of Run 2 (5 voters, 3 rounds)

Verification of Run 2 (5 voters, 3 rounds)

The majority result of each round can be checked against the printed votes:

- **Round 1:** YES, NO, YES, NO, YES gives 3 YES and 2 NO, so the result is YES.
- **Round 2:** NO, YES, YES, NO, YES gives 3 YES and 2 NO, so the result is YES.
- **Round 3:** YES, YES, NO, NO, NO gives 2 YES and 3 NO, so the result is NO.

Once again, the output is not in strict round order. Voters 0 to 3 printed all three of their votes before the judge printed the Round 1 header, and Voter 4's Round 2 vote appears before the Round 1 result. This happens because each voter runs independently and does not wait for the judge, and the judge only announces a round after it has read every voter's vote for that round.